Software Requirements Specification (SRS)

Perishables Management System

Project	**Perishables Management System (PMS)**
Version	1.0
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Date	16 October 2026
Status	Draft — partially implemented; verify against repository

Revision History

Version	Date	Author	Change summary
1.0	28 October 2026	Four-member Software Engineering project team	Initial project-specific draft

Approvals

Role	Name	Date
Project team representative	Chiranth J	16th October

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1. Introduction

1.1 Purpose

This SRS defines the functional and non-functional requirements for the Perishables Management System (PMS), an application intended to help retail stores manage perishable inventory such as fresh fruits and vegetables. It is the reference for implementation, testing, and project evaluation.

1.2 Scope

PMS supports recording stock batches, tracking quantities and expiry dates, identifying items approaching expiry, monitoring stock levels, and providing reports that help staff reduce avoidable spoilage. The initial project is a software prototype for a retail store. It does not include automated sensor hardware, supplier payment processing, or integration with external enterprise systems unless added and approved later.

1.3 Audience

Student developers, testers, store staff acting as users, project evaluators, and the teaching assistant.

1.4 Definitions

Term	Definition
PMS	Perishables Management System
Batch	A quantity of one product received together, with its own received and expiry dates
Expiry alert	A notification or visual warning for a batch nearing or past its expiry date
FEFO	First Expired, First Out; prioritize stock with the earliest expiry date
FR / NFR	Functional Requirement / Non-functional Requirement
RBAC	Role-Based Access Control

2. Overall Description

2.1 Product Perspective

PMS is a store-level application with a user interface, application/service layer, and persistent database. Users enter product and batch details; the application validates and stores them, calculates stock/expiry status, and presents dashboards and reports.

2.2 Major Product Functions

- Authenticate users and apply role-based permissions.
- Create, view, update, and deactivate product records.
- Record incoming stock as batches with quantity, received date, expiry date, and optional supplier/reference.
- Adjust stock for sales, waste/spoilage, returns, and corrections while preserving an audit trail.
- Show low-stock, near-expiry, and expired-batch alerts.
- Search and filter inventory by product, category, quantity status, and expiry status.
- Provide FEFO-oriented stock handling guidance.
- Generate inventory, wastage, and expiry reports.

2.3 User Roles and Characteristics

Role	Needs / permissions
Administrator	Manage users and roles; configure alert thresholds; view all reports; maintain reference data.
Store manager	Manage products and batches; review alerts; approve or inspect stock adjustments; view reports.
Inventory staff	Record stock received, update quantities, record waste, and search inventory.
Viewer / evaluator	Read dashboards and reports without changing inventory.

2.4 Operating Environment

The prototype is intended to run on a modern desktop or laptop browser, with a Python-based application/service layer and a relational database as selected by the team. The application should support a local development environment and a reproducible setup documented in the repository.

2.5 Constraints and Assumptions

- Users enter product and expiry information accurately; the prototype does not independently verify physical stock.
- A batch may be tracked separately from other batches of the same product.
- Alert thresholds are configurable or use documented defaults.
- The system is for one store in the initial release unless multi-store support is explicitly implemented.
- All date comparisons use the store's configured local date/time.
- The implementation and selected technologies must match the actual repository; record deviations in the final demo/documentation.

3. External Interface Requirements

3.1 User Interface

- Dashboard showing total active products/batches, low-stock items, near-expiry batches, and expired stock.
- Forms with required-field validation and clear error messages.
- Inventory table with search, sort, and filters.
- Reports readable on a standard laptop screen; important statuses must not rely on colour alone.

3.2 Software Interfaces

- Application services expose operations for product, batch, stock-adjustment, alert, and report workflows.
- The database stores users, products, batches, stock movements, and audit events.
- Optional email/SMS notifications are outside the initial scope unless implemented and documented.

3.3 Communications

For local development, browser-to-server communication may use localhost. Any deployed environment should use HTTPS, protect credentials, and avoid exposing the database directly to end users.

4. System Features / Functional Requirements

Req ID	Requirement (shall)	Priority	Acceptance criteria
PMS-F-001	The system shall allow an authorized user to sign in and sign out.	High	Valid credentials create a session; invalid credentials are rejected.
PMS-F-002	The system shall enforce role-based permissions for administrative and inventory actions.	High	A user without permission cannot access or execute a restricted action.
PMS-F-003	The system shall allow authorized users to create a product with name, category, unit, and applicable minimum-stock threshold.	High	A valid product is saved and appears in inventory; required fields are validated.
PMS-F-004	The system shall allow authorized users to view, edit, search, and filter product records.	High	Changes persist and search/filter results match the selected criteria.
PMS-F-005	The system shall allow users to record an incoming stock batch linked to a product.	High	Batch quantity, received date, and expiry date are stored after validation.
PMS-F-006	The system shall validate that batch quantity is positive and expiry date is not earlier than received date.	High	Invalid quantities or date ranges are rejected with a clear message.
PMS-F-007	The system shall calculate available stock from recorded stock movements or batch quantities.	High	Displayed stock matches the sum of valid receipts and adjustments.
PMS-F-008	The system shall record stock reductions for sale/use, waste, or spoilage with reason and quantity.	High	A valid reduction updates available stock and records the reason.
PMS-F-009	The system shall prevent stock reductions that exceed the available quantity unless an authorized correction workflow is used.	High	Over-reduction is rejected or requires the defined privileged correction flow.
PMS-F-010	The system shall identify batches that are within the configured near-expiry window.	High	A batch with expiry date inside the window appears as near expiry.
PMS-F-011	The system shall identify batches whose expiry date has passed and whose remaining quantity is positive.	High	Expired stock is visibly flagged and included in the expired-stock view.
PMS-F-012	The system shall identify products whose available quantity is below the configured minimum-stock threshold.	Medium	Low-stock products appear in the low-stock view.
PMS-F-013	The system shall allow users to record expired or spoiled quantities as waste and retain the transaction history.	High	Waste is reflected in stock and wastage records.
PMS-F-014	The system shall provide FEFO-oriented sorting or guidance for batches of the same product.	Medium	The earliest valid expiry batch is listed first when FEFO sorting is selected.
PMS-F-015	The system shall generate inventory reports for a selected date or current snapshot.	Medium	Report totals reconcile with the inventory data for the selected scope.
PMS-F-016	The system shall generate a wastage report grouped by date, product, and recorded reason.	Medium	The report includes the selected period and corresponding waste quantities.
PMS-F-017	The system shall provide a near-expiry and expired-stock report.	High	The report includes product, batch, expiry date, and remaining quantity.
PMS-F-018	The system shall record an audit event for important changes to products, batches, users, and stock quantities.	High	Audit entries identify action, timestamp, and acting user where available.
PMS-F-019	The system shall allow an administrator to configure the near-expiry window and low-stock thresholds.	Medium	New settings affect subsequent alert calculations.
PMS-F-020	The system shall display validation and operation errors without exposing stack traces or sensitive data.	High	User sees a useful message while technical details remain in protected logs.

5. Non-functional Requirements

Req ID	Requirement	Category	Acceptance / measurement
PMS-NF-001	For a dataset of up to 5,000 active batches on the project test machine, common inventory searches and dashboard loads should complete within 2 seconds for at least 90% of measured requests.	Performance	Record timings across at least 20 repeated requests; 90% meet target.
PMS-NF-002	The application shall preserve committed product, batch, and stock-movement data across normal application restarts.	Reliability	Restart application and verify previously committed records remain intact.
PMS-NF-003	Passwords shall not be stored in plaintext; access to protected actions shall require authentication and authorization.	Security	Inspect stored user data and test unauthorized requests.
PMS-NF-004	The user interface shall provide consistent navigation, readable labels, validation feedback, and keyboard-operable core forms where supported by the framework.	Usability / Accessibility	Complete defined usability checklist for product, batch, and stock workflows.
PMS-NF-005	The codebase shall separate UI/ routes, business logic/services, and data persistence sufficiently to support unit testing and maintenance.	Maintainability	Core business rules can be unit-tested without launching the full UI.
PMS-NF-006	The application shall provide a documented setup procedure and run on the team's supported development environment.	Portability	A team member follows README setup steps on a clean environment and launches the application.

6. Security Objectives and Requirements

6.1 Security Objectives

- Protect inventory and user-account data from unauthorized viewing or modification.
- Preserve the integrity and traceability of stock movements so quantity changes can be investigated.

6.2 Security Requirements

Req ID	Requirement	Priority	Acceptance / test
PMS-SR-001	The system shall require authentication before allowing inventory changes.	High	Unauthenticated requests to protected operations are rejected.
PMS-SR-002	The system shall enforce role-based authorization on the server/service layer, not only by hiding UI controls.	High	Direct requests by a role without permission are denied.
PMS-SR-003	The system shall store passwords using a suitable salted password-hashing mechanism and shall never log passwords.	High	Storage and log inspection show no plaintext passwords.
PMS-SR-004	The system shall validate and normalize user-supplied values on the server side.	High	Malformed, missing, negative, or out-of-range values are rejected safely.
PMS-SR-005	The system shall record audit events for stock adjustments and privileged changes without including passwords or other secrets.	High	Audit history records action and actor while excluding secret values.

7. Quality Attributes and Acceptance Tests

Acceptance exit criteria: all high-priority functional requirements have been implemented and tested; no open critical defect remains; stock totals reconcile; and each implemented requirement is linked to at least one test case in the traceability matrix.

Quality attribute	Acceptance test
Correctness	Reconcile batch quantities against receipt, reduction, and waste movements.
Security	Test sign-in, role restrictions, validation, and audit-log content.
Performance	Measure dashboard and inventory search against PMS-NF-001.
Usability	Ask a tester to complete product creation, batch receipt, stock reduction, and report viewing using the UI alone.

Maintainability	Run unit tests for business services and verify test results are repeatable.
Portability	Follow documented setup steps in a clean virtual environment.

8. UML Use-Case Models

Use-case diagram 1 — Inventory operations. Actors: Inventory Staff and Store Manager. Use cases: Sign in; Search/view inventory; Add product; Record incoming batch; Record stock reduction; Record waste; View near-expiry/expired alerts. Store Manager can additionally review reports and approve privileged corrections.

Use-case diagram 2 — Administration and reporting. Actors: Administrator and Store Manager. Use cases: Manage users/roles; Configure alert thresholds; View inventory report; View wastage report; View expiry report; Review audit history. Inventory Staff may view alerts but cannot manage users or change system-wide settings.

9. Requirements Traceability Matrix (RTM)

Req ID	Short requirement	Module	Test case(s)	Status
PMS-F-001	Sign in/out	Authentication	TC-AUTH-01, TC-AUTH-02	Not tested
PMS-F-003	Create product	Product service/UI	TC-PROD-01, TC-PROD-02	Not tested
PMS-F-005	Record incoming batch	Inventory service	TC-BATCH-01, TC-BATCH-02	Not tested
PMS-F-006	Validate quantity/dates	Inventory service	TC-VAL-01, TC-VAL-02	Not tested
PMS-F-008	Record stock reduction	Stock service	TC-STOCK-01, TC-STOCK-02	Not tested
PMS-F-010	Near-expiry detection	Alert service	TC-ALERT-01	Not tested
PMS-F-011	Expired-stock detection	Alert service	TC-ALERT-02	Not tested
PMS-F-015	Inventory report	Reporting module	TC-REP-01	Not tested
PMS-F-018	Audit events	Audit service	TC-AUDIT-01	Not tested
PMS-NF-001	Performance target	Application/UI	TC-PERF-01	Not tested
PMS-NF-003	Secure credentials	Authentication	TC-SEC-01	Not tested
PMS-SR-002	Role authorization	Authorization layer	TC-SEC-02	Not tested